

Performance Comparison of Session-based Recommendation Algorithms based on GNNs

Faisal Shehzad, Dietmar Jannach

Faisal.Shehzad@aau.at, Dietmar.Jannach@aau.at

Introduction and Motivation

Recommender systems

Conventional recommender systems

Session based recommender systems
(SBRs)

❑ From a technical point of view

1. Most SBRs are based on neural models
2. Need user-item interaction data to generate a recommendation list

❑ Quite surprisingly

1. Many studies [1, 2] have shown that simple models (kNN, etc.) are competitive
2. Various factors [3] contribute to this “Phantom progress” such as compared to one family of algorithms (neural models), not properly tune baseline models, etc.

❑ An emerging trends in SBRs (graph neural networks)

Research Question

- ❑ Although we know that simple models outperform the complex models in majority of cases. However, we want to know to what extent this is true for GNN models



Results

❑ RQ1: to what extent simple models outperform GNN models

Accuracy measures

Table 1: Best performing model for each dataset and each accuracy measure				
Datasets	RSC15 (1/64)	RSC15 (1/12)	RETAIL	DIGI
MRR@10	STAN	SR	VSTAN	SFSKNN
MRR@20	STAN	SR	VSTAN	SFSKNN
HR@10	CM-HGNN	VSTAN	VSTAN	SR-GNN
HR@20	CM-HGNN	GNRRW	VSTAN	SR-GNN

Note: GNN models are highlighted in blue color

❑ Findings

1. Simple models outperform all GNN models in case of MRR values on all datasets (MRR@20 was our optimization criterion)
2. In six cases, GNN models achieve the best HR@K
3. VSTAN and STAN consistently give competitive results for all datasets
4. More than a half decade, there have been strong baselines with more than 100 citations. However, no one considers them. why?
5. Empirical studies show the effectiveness of simple models (offline settings)

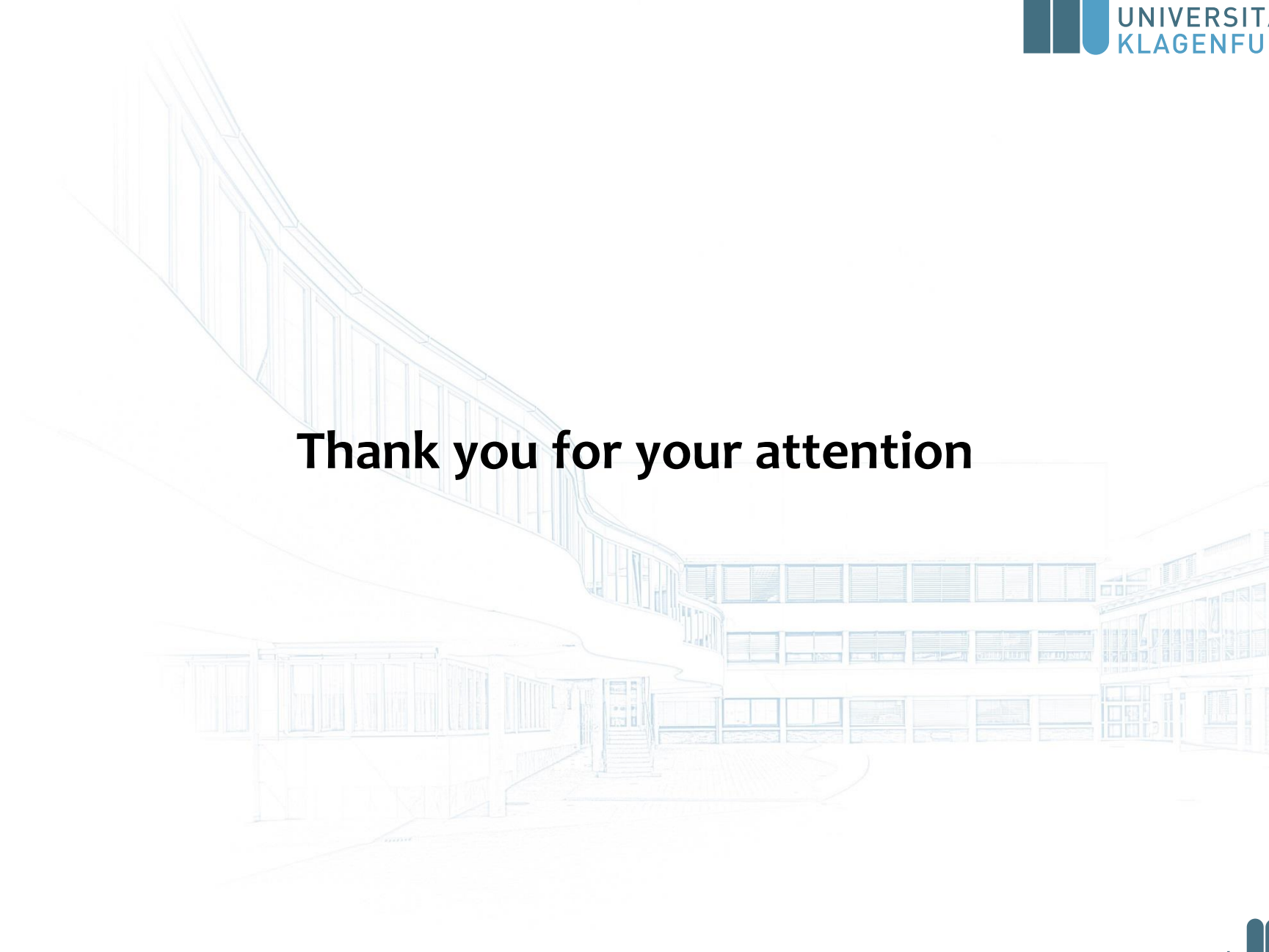
Results

- ❑ Additional experiments regarding the bad research practices in the examined literature

Sr. No.	Bad Research Practices	Performance difference between the worst and the best MRR value of a model
1	Embedding size	> 40%
2	Random seeds	> 30%
3	Models' tuning on test data	3% to 4%
4	Comparison with non-tuned models	Everyone's a Winner! On Hyperparameter Tuning [12]

Conclusions

- ❑ Analysis of identified articles shows that new proposals were compared to
 1. With only recent algorithms
 2. One family of algorithms (neural models)
 3. Were not compared to simpler models that show competitive performance in empirical studies
- ❑ On a positive note, there is much room for improvement in the SBRS as there are not enough studies that take into account additional information of items. So, there is “still a long way to go”



Thank you for your attention

References

1. Garg, Diksha, et al. "Sequence and Time Aware Neighborhood for Session-based Recommendations: STAN." Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval. 2019.
2. Jannach, Dietmar, and Malte Ludewig. "When Recurrent Neural Networks Meet the Neighborhood for Session-based Recommendation." Proceedings of the Eleventh ACM Conference on Recommender Systems. 2017.
3. Cremonesi, Paolo, and Dietmar Jannach. "Progress in Recommender Systems Research: Crisis? What crisis?." AI Magazine 42.3 (2021): 43-54.
4. Wu, Shu, et al. "Session-based Recommendation with Graph Neural Networks." Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 33. No. 01. 2019.
5. Wang, Ziyang, et al. "Global Context Enhanced Graph Neural Networks for Session-based Recommendation." Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. 2020.
6. Xia, Xin, et al. "Self-supervised Graph Co-training for Session-based Recommendation." Proceedings of the 30th ACM International Conference on Information & Knowledge Management. 2021.
7. Zhang, Zizhuo, and Bang Wang. "Graph Neighborhood Routing and Random Walk for Session-based Recommendation." 2021 IEEE International Conference on Data Mining (ICDM). IEEE, 2021.
8. Zhang, Zizhuo, and Bang Wang. "Fusion of Latent Categorical Prediction and Sequential Prediction for Session-based recommendation." Information Sciences 569 (2021): 125-137.
9. Xu, Hao, et al. "Category-aware Multi-relation Heterogeneous Graph Neural Networks for Session-based Recommendation." Knowledge-Based Systems 251 (2022): 109246.
10. Yu, Feng, et al. "TAGNN: Target Attentive Graph Neural Networks for Session-based Recommendation." Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. 2020.
11. Lai, Siqi, et al. "An Attribute-driven Mirror Graph Network for Session-based Recommendation." Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval. 2022.
12. Shehzad, Faisal, and Dietmar Jannach. "Everyone's a Winner! on Hyperparameter Tuning of Recommendation Models." Proceedings of the 17th ACM Conference on Recommender Systems. 2023.